

Bluestock MF Analytics

Mutual Fund Industry Analysis

Comprehensive Capstone Project Report

Submitted by:	Vaishnavi Rao
Project:	Bluestock Fintech MF Analytics Capstone
Date:	June 2026
Tools:	Python, Pandas, SQLite, Power BI, ReportLab
GitHub:	github.com/Vaishnavi-Rao-06/Fintech_Capstone_Project

Table of Contents

1. Executive Summary
2. Data Sources & Description
3. ETL Pipeline
4. EDA Findings
5. Performance Analysis
6. Dashboard Screenshots
7. Advanced Analytics
8. Recommendations
9. Limitations

1. Executive Summary

This report presents a comprehensive analysis of the Indian Mutual Fund industry using data sourced from AMFI and mfapi.in. The Bluestock MF Analytics project covers 40 mutual fund schemes across 10 major fund houses, analyzing NAV trends from January 2022 to May 2026.

The project involved building a complete data engineering and analytics pipeline — from raw data ingestion to interactive Power BI dashboards. Key highlights include:

- SBI Mutual Fund dominates the industry with the highest AUM share
- Monthly SIP inflows reached an all-time high of ₹31,002 Crore in December 2025
- Total industry folios doubled from 13.26 Crore to 26.12 Crore between 2022-2025
- Mirae Asset Large Cap Fund leads risk-adjusted returns with Sharpe ratio of 1.45
- Small cap funds carry highest daily risk with VaR exceeding 2.5%
- B30 cities represent a significant untapped growth opportunity

2. Data Sources & Description

The project utilized 10 structured datasets sourced from AMFI (Association of Mutual Funds in India) and live NAV data fetched via the mfapi.in REST API. All datasets cover the period January 2022 to December 2025.

#	Dataset	Rows	Key Columns	Source
1	Fund Master	40	amfi_code, fund_house, category	AMFI
2	NAV History	46,000	amfi_code, date, nav	mfapi.in
3	AUM by Fund House	160	date, fund_house, aum_crore	AMFI
4	Monthly SIP Inflows	48	month, sip_inflow_crore	AMFI
5	Category Inflows	576	month, category, net_inflow	AMFI
6	Folio Count	48	month, total_folios_crore	AMFI
7	Scheme Performance	40	sharpe, alpha, beta, returns	AMFI
8	Investor Transactions	50,000+	investor_id, amount, type	Simulated
9	Portfolio Holdings	1,000+	amfi_code, sector, weight	AMFI
10	Benchmark Indices	5,000+	date, index_name, close	NSE/BSE

3. ETL Pipeline

3.1 Extract

Data was extracted from two sources: mfapi.in REST API for live NAV data and 10 pre-compiled CSV datasets from AMFI. Python's requests library was used to fetch JSON responses from the API, which were then parsed and saved as raw CSVs.

3.2 Transform

- Date parsing: All date columns converted to datetime using `pd.to_datetime(format='mixed')`
- Forward fill: Missing NAV values for weekends/holidays filled using business day resampling (`resample('B').ffill()`)
- Duplicate removal: `drop_duplicates()` applied on `amfi_code` + date combinations
- Validation: NAV > 0, expense ratio 0.1%-2.5%, `amount_inr` > 0
- Standardization: `transaction_type` values standardized to SIP/Lumpsum/Redemption
- Negative Sharpe flagging: `negative_sharpe_flag` column added to performance data

3.3 Load

All 10 cleaned datasets were loaded into SQLite database (`bluestock_mf.db`) using SQLAlchemy's `create_engine` and `DataFrame.to_sql()` method. The database contains 4 core tables: `dim_fund`, `fact_nav`, `fact_transactions`, `fact_performance`, plus 6 supporting tables for AUM, SIP, folios, categories, holdings, and benchmarks.

4. EDA Findings

4.1 NAV Growth Trends

HDFC Short Term Debt Fund showed the highest NAV growth from ■3,200 to ■4,500+ between 2022-2026, significantly outperforming peers. The 2023 bull run and 2024 market corrections are clearly visible in the NAV trend charts.

4.2 AUM Dominance

SBI Mutual Fund dominates with ■12.5L Crore AUM, nearly double the next competitor ICICI Prudential MF. The top 3 fund houses (SBI, ICICI, HDFC) collectively manage over 50% of industry AUM.

4.3 SIP Growth Story

Monthly SIP inflows showed consistent growth from ■12,000 Crore in January 2022 to ■31,002 Crore in December 2025 — a 158% increase. This reflects growing retail investor confidence in systematic investing.

4.4 Category Analysis

Liquid funds consistently show the highest net inflows across all months, acting as parking vehicles for short-term money. Equity categories show seasonal patterns with higher inflows during market corrections.

4.5 Investor Demographics

The 25-35 age group dominates investor distribution, indicating strong millennial participation. Male investors account for 65% of transactions, highlighting a significant gender gap in MF participation.

4.6 Geographic Distribution

T30 cities account for over 70% of total SIP investments, with Maharashtra and Delhi leading in absolute transaction volumes. B30 cities show growing but still low participation.

4.7 Folio Growth

Total folios nearly doubled from 13.26 Crore in January 2022 to 26.12 Crore in December 2025, reflecting massive mutual fund adoption in India.

4.8 Correlation Analysis

Top 10 equity funds show high positive correlation (>0.85), suggesting they move together during market events. This limits diversification benefits within the equity category.

5. Performance Analysis

5.1 CAGR Analysis

Compound Annual Growth Rate was computed for 1-year, 3-year, and 5-year periods using the formula: $CAGR = (NAV_end / NAV_start)^{(1/n)} - 1$. Small cap and mid cap funds consistently showed higher CAGRs (18-22%) compared to debt funds (7-8%).

Top 5 Funds by 3-Year CAGR:

Scheme Name	Fund House	1yr CAGR	3yr CAGR	5yr CAGR
Axis Midcap Fund - Regular - G	Axis Mutual Fun	22.26%	35.11%	24.45%
Mirae Asset Large Cap Fund - R	Mirae Asset MF	20.36%	34.00%	26.80%
ICICI Pru Bluechip Fund - Dire	ICICI Prudentia	13.06%	32.49%	20.23%
HDFC Mid-Cap Opportunities Fun	HDFC Mutual Fun	53.23%	32.44%	26.07%
ICICI Pru Midcap Fund - Regula	ICICI Prudentia	29.60%	31.78%	28.38%

5.2 Risk Metrics

Risk metrics were computed using daily returns derived from NAV history. Risk-free rate used: 6.5% (RBI repo rate proxy). All metrics annualized using 252 trading days.

Top 5 Funds by Sharpe Ratio:

Scheme Name	Fund House	Sharpe Ratio
Mirae Asset Large Cap Fund - Regula	Mirae Asset MF	1.4483
Kotak Flexicap Fund - Regular - Gro	Kotak Mahindra MF	1.3067
Mirae Asset Tax Saver Fund - Regula	Mirae Asset MF	1.2349
SBI Bluechip Fund - Regular Plan -	SBI Mutual Fund	1.2083
ICICI Pru Midcap Fund - Regular - G	ICICI Prudential MF	1.1801

5.3 VaR & CVaR Analysis

Historical Value at Risk (VaR) at 95% confidence level was computed as the 5th percentile of daily return distribution. Conditional VaR (CVaR) represents the expected loss beyond the VaR threshold.

Top 5 Highest Risk Funds (by VaR):

Scheme Name	Fund House	VaR 95%	CVaR 95%
SBI Small Cap Fund - Direct PI	SBI Mutual Fund	-2.6859%	-3.2384%
Axis Small Cap Fund - Regular	Axis Mutual Fun	-2.6188%	-3.1667%
ABSL Small Cap Fund - Regular	Aditya Birla Su	-2.6021%	-3.2459%

Nippon India Small Cap Fund -	Nippon India MF	-2.5438%	-3.2304%
SBI Small Cap Fund - Regular P	SBI Mutual Fund	-2.4507%	-3.0595%

5.4 Fund Scorecard

A composite fund scorecard was built using weighted ranks of 5 metrics: $30\% \times 3\text{yr Return} + 25\% \times \text{Sharpe} + 20\% \times \text{Alpha} + 15\% \times \text{Expense Ratio (inverse)} + 10\% \times \text{Max Drawdown (inverse)}$

Scheme Name	Fund House	3yr CAGR	Sharpe	Score
ICICI Pru Midcap Fund - Regu	ICICI Prudentia	31.78%	1.18	84.50
Axis Midcap Fund - Regular -	Axis Mutual Fun	35.11%	1.00	80.75
HDFC Mid-Cap Opportunities F	HDFC Mutual Fun	32.44%	1.09	80.50
Mirae Asset Large Cap Fund -	Mirae Asset MF	34.00%	1.45	80.00
Kotak Flexicap Fund - Regula	Kotak Mahindra	29.58%	1.31	78.25

6. Dashboard Screenshots

Page 1: Industry Overview — Screenshot not found

Page 2: Fund Performance — Screenshot not found

Page 3: Investor Analytics — Screenshot not found

Page 4: SIP & Market Trends — Screenshot not found

7. Advanced Analytics

7.1 VaR & CVaR

Historical VaR at 95% confidence computed for all 40 funds. Small cap and mid cap funds show highest VaR, confirming higher risk profiles.

7.2 Rolling Sharpe Ratio

90-day rolling Sharpe ratio computed for top 5 funds. Clear dip visible during 2024 market correction period.

7.3 Investor Cohort Analysis

Investors grouped by first transaction year. 2024 cohort shows higher average SIP amounts compared to 2025 cohort.

7.4 SIP Continuation Analysis

Investors with 6+ SIP transactions analyzed for gaps. At-risk investors (gap >35 days) identified for proactive engagement.

7.5 Fund Recommender

Rule-based recommender maps investor risk appetite (Low/Moderate/High) to top 3 funds by Sharpe ratio within matching risk category.

7.6 Sector HHI Analysis

Herfindahl-Hirschman Index computed for each equity fund. Funds with HHI >0.25 flagged as highly concentrated in Financial Services sector.

8. Recommendations

8.1 For Conservative Investors

ICICI Pru Liquid Fund offers the best Sharpe ratio (0.50) among low-risk funds. Suitable for short-term parking of funds with capital preservation.

8.2 For Moderate Risk Investors

Mirae Asset Large Cap Fund leads with Sharpe ratio of 1.45 and consistent 3-year CAGR. SBI Bluechip and Kotak Flexicap are strong alternatives.

8.3 For Aggressive Investors

Mirae Asset Tax Saver Fund and ICICI Pru Midcap Fund offer best risk-adjusted returns for high-risk appetite investors.

8.4 For Fund Houses

Target B30 cities for SIP growth campaigns — currently underrepresented at 30% of transactions despite large population base.

8.5 For Retention

Flag at-risk SIP investors with gaps >35 days and initiate proactive engagement through personalized communications.

8.6 For Portfolio Construction

Avoid funds with high HHI scores (>0.25) for better sector diversification. Mix large cap, mid cap and debt funds for optimal risk-return balance.

9. Limitations

9.1 Simulated Transaction Data

Investor transaction data is synthetically generated and may not accurately reflect real investor behavior, geographic distribution, or demographic patterns.

9.2 Limited Fund Universe

Analysis covers only 40 fund schemes. The actual AMFI universe contains 1,900+ schemes. Broader analysis would provide more representative insights.

9.3 Benchmark Limitations

Fund performance is compared only against Nifty 100. Sector funds and debt funds require category-specific benchmarks for fair comparison.

9.4 VaR Methodology

Historical VaR assumes future returns follow past distributions. Monte Carlo simulation would provide more robust risk estimates.

9.5 Static Dashboard

Power BI dashboard is published locally and requires manual data refresh. Real-time updates need cloud deployment with scheduled refresh.

9.6 Recommendation Model

The fund recommender is rule-based and uses only Sharpe ratio. A more sophisticated model would incorporate investor horizon, tax status, and goals.